The Hypercontractivity Constant is the Largest Tensorized SDPI Constant for a Bivariate Source with Binary and Finite Components

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Abstract

In this paper, we prove that for a bivariate source (X, Y) where the two components take binary and finite values, respectively, the hypercontractivity constant is the maximum among all Φ-strong data processing inequality (Φ-SDPI) constants for twice differentiable, convex Φ such that $\frac{1}{\Phi''}$ is concave. This class of functions Φ's was widely considered in the literature to ensure the tensorization of Φ-SDPI constants.

Index Terms

Strong data processing inequality, hypercontractivity inequality, tensorization, subadditivity.

I. Introduction

Let (X, Y) be a pair of random variables defined on the product of finite alphabets $\mathcal{X} \times \mathcal{Y}$. Let $f : \mathcal{X} \to \mathbb{R}_{\geq 0}$, and $f(X)$ a random variable induced by X. Let $\Phi : \mathbb{R}_{\geq 0} \to \mathbb{R}$. Then the Φ-entropy of f is defined by

$$H_\Phi(f) = \mathbb{E}[\Phi(f)] - \Phi(\mathbb{E}f).$$

In this paper we always assume that Φ is convex, in which case

$$H_\Phi(f) \geq 0,$$

by Jensen's inequality. For the choice of $\Phi(t) = t^2$, the Φ-entropy reduces to variance: $H_\Phi(f) = \mathrm{Var}(f)$.

Let $\mathcal{F}$ be the class of all twice differentiable functions $\Phi : \mathbb{R}_{\geq 0} \to \mathbb{R}$ such that $\Phi'' > 0$ (i.e., Φ is convex) and $\frac{1}{\Phi''}$ is concave. Let $\widetilde{\mathcal{F}}$ be the class of all non-constant functions $f : \mathcal{X} \to \mathbb{R}_{\geq 0}$ such that $\mathbb{E}[f] = 1$.

Definition 1. For a function $\Phi \in \mathcal{F}$, define the Φ-strong data processing inequality constant (Φ-SDPI constant or Φ-entropy contraction coefficient) $\eta_\Phi(X, Y)$ as

$$\eta_\Phi(X, Y) := \sup_{f \in \widetilde{\mathcal{F}}} \frac{H_\Phi(\mathbb{E}[f|Y])}{H_\Phi(f)}.$$

The Φ-SDPI constant was recently systematically studied in [1], [2] for discrete distributions and in [3], [4] for continuous distributions. The Φ-SDPI constant was first discovered by Ahlswede and GÃ¡cs in a landmark paper [5], in which a special Φ-entropy, the relative entropy, was considered. For this case, the Φ-SDPI constant is also known to be equal to the hypercontractivity constant. The latter one is defined as follows.

Definition 2. For $p \geq 1$ define

$$q_p^*(X, Y) := \inf \left\{ q : \|\mathbb{E}[f(X)|Y]\|_p \leq \|f(X)\|_q, \forall f : \mathcal{X} \to \mathbb{R}_{\geq 0} \right\}.$$

Moreover, define the hypercontractivity constant $s^*(X, Y)$ as

$$s^*(X, Y) := \lim_{p \to \infty} \frac{q_p^*(X, Y) - 1}{p - 1}.$$

For Φ-entropy, it is usually assumed that Φ is convex. However, when the tensorization property is required, then it is usual to restrict $\Phi \in \mathcal{F}$ [1]. In fact, assume Φ is twice differentiable with $\Phi'' > 0$, then the condition $\Phi \in \mathcal{F}$ is necessary and sufficient for Φ-entropy to be subadditive [6, Theorem 14.1 and Proposition 14.3]. Subadditivity of Φ-entropy implies the tensorization property of Φ-SDPI constant [1]. However, it is unclear whether the converse holds or not. Throughout this paper, we focus on the case $\Phi \in \mathcal{F}$, i.e., we focus on the Φ-entropy which satisfies the tensorization property.

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From the convexity of Φ, $\eta_\Phi(X,Y) \in [0,1]$. Moreover, if X and Y are independent, then $\eta_\Phi(X,Y) = 0$. If X and Y have common randomness (i.e., $f(X) = g(Y)$ for some non-constant f and g), then $\eta_\Phi(X,Y) = 1$. The doubly symmetric binary source (DSBS) with correlation ρ, denoted by DSBS(ρ), is defined as a pair of binary random variables (X,Y) such that

$$\mathbb{P}(X=0,Y=0) = \mathbb{P}(X=1,Y=1) = \frac{1+\rho}{4},$$
$$\mathbb{P}(X=0,Y=1) = \mathbb{P}(X=1,Y=0) = \frac{1-\rho}{4}.$$

It is known that [2] for the DSBS, $\eta_\Phi(X,Y) = \rho^2$ holds for all $\Phi \in \mathcal{F}$.

When $\Phi(t) = t^2$, $\eta_\Phi(X,Y)$ is exactly $\rho_{\mathrm{m}}^2(X,Y)$ as shown in [7], where $\rho_{\mathrm{m}}(X,Y)$ denotes the maximal correlation between X,Y. Moreover, for the choice of $\Phi(t) = t\log t$, $\eta_\Phi(X,Y)$ is known to be equal to the hypercontractivity constant $s^*(X,Y)$ [5]. Hence the maximal correlation and the hypercontractivity constant can be seen as special cases of SDPI constants. For all these constants, they all satisfy the following data processing inequality: for any $U \to X \to Y \to V$,

$$\eta_\Phi(U,V) \le \eta_\Phi(X,Y) \tag{1}$$

for any convex Φ. The hypercontractivity inequality has been widely applied in various fields, including discrete Fourier analysis, combinatorics, probability theory, and information theory; please refer to [5], [8]–[11]. A natural question arises: Can we apply the Φ-SDPI to strengthen the hypercontractivity inequality by finding a better Φ? In this paper, we show that when (X,Y) is the DSBS and (U,V) is a pair of binary random variables, the hypercontractivity inequality is the strongest Φ-SDPI. We next state our results.

WLOG, one can restrict Φ to be a function such that

$$\Phi(1) = 0 \tag{2}$$
$$\Phi'(1) = 0 \tag{3}$$
$$\Phi''(1) = 1, \tag{4}$$

since for fixed f, $\frac{H_\Phi(\mathbb{E}[f|Y])}{H_\Phi(f)}$ does not change if we replace $\Phi(t)$ with $\alpha\Phi(t)+\beta t+\gamma$ for $\alpha > 0, \beta, \gamma \in \mathbb{R}$. Under the conditions (2)-(4), $\Phi \in \mathcal{F}$ is uniquely determined by $\frac{1}{\Phi''}$. Let $\mathcal{G}$ be the class of all concave functions $g : \mathbb{R}_{\ge 0} \to \mathbb{R}_{>0}$. Equivalent definitions of $\mathcal{G}$ can be found in [6, Exercise 14.2]. By definition, we know that any $g \in \mathcal{G}$ is nondecreasing, and moreover,

$$\mathcal{F} = \left\{ r \in \mathbb{R}_{\ge 0} \mapsto \int_1^r \int_1^s \frac{1}{g(t)} dt ds \in \mathbb{R}_{\ge 0} : g \in \mathcal{G} \right\}.$$

Without ambiguity, we also denote $H_\Phi(f)$ as $H_g(f)$. We next provide several properties of sets $\mathcal{F}$ and $\mathcal{G}$, the function Φ, and the Φ-SDPI constant.

Lemma 1. *Both $\mathcal{F}$ and $\mathcal{G}$ are convex.*

Proof: For $\Phi_1, \Phi_2 \in \mathcal{F}$ and $0 \le \lambda \le 1$, denote $\Psi_\lambda = \lambda\Phi_1 + (1-\lambda)\Phi_2$. Then it is easy to verify $\Psi_\lambda'' > 0$. Hence it remains to prove that $\frac{1}{\Psi_\lambda''}$ is concave. For a twice differentiable function Φ, $\frac{1}{\Phi''}$ is concave if and only if $(s,t) \mapsto p\Phi(s) + (1-p)\Phi(t) - \Phi(ps + (1-p)t)$, for any $p \in [0,1]$, is jointly convex. Hence it follows that $\frac{1}{\Psi_\lambda''}$ is concave. ■

Lemma 2. *Given f, $\Phi \in \mathcal{F} \mapsto \frac{H_\Phi(\mathbb{E}[f|Y])}{H_\Phi(f)} \in \mathbb{R}_{>0}$ is a quasiconvex function.*

Proof: By definition, H_Φ is a linear functional in Φ. That is, for $\Psi = \lambda\Phi_1 + (1-\lambda)\Phi_2$,

$$H_\Psi(f) = \lambda H_{\Phi_1}(f) + (1-\lambda) H_{\Phi_2}(f).$$

Hence

$$\begin{aligned}\frac{H_\Psi(\mathbb{E}[f|Y])}{H_\Psi(f)} &= \frac{\lambda H_{\Phi_1}(\mathbb{E}[f|Y]) + (1-\lambda) H_{\Phi_2}(\mathbb{E}[f|Y])}{\lambda H_{\Phi_1}(f) + (1-\lambda) H_{\Phi_2}(f)} \\ &\le \max\left\{ \frac{H_{\Phi_1}(\mathbb{E}[f|Y])}{H_{\Phi_1}(f)}, \frac{H_{\Phi_2}(\mathbb{E}[f|Y])}{H_{\Phi_2}(f)} \right\}\end{aligned}$$

That is, given f, $\Phi \in \mathcal{F} \mapsto \frac{H_\Phi(\mathbb{E}[f|Y])}{H_\Phi(f)} \in \mathbb{R}_{>0}$ is a quasiconvex function. ■

Lemma 3. *Given f, $\sup_{\Phi\in\mathcal{F}} \frac{H_\Phi(\mathbb{E}[f|Y])}{H_\Phi(f)}$ is attained by some Φ.*

The proof of Lemma 3 is provided in Appendix A.

It is well-known that $\rho_{\mathrm{m}}^2(X,Y) \le s^*(X,Y)$, where $\rho_{\mathrm{m}}(X,Y)$ is the maximal correlation. Moreover, Raginsky [1] proved that $\rho_{\mathrm{m}}^2(X,Y) \le \eta_\Phi(X,Y)$ for any $\Phi \in \mathcal{F}$. The following theorem gives the relationship between $\eta_\Phi(X,Y)$ and $s^*(X,Y)$ for binary X.

Theorem 1. *If X is binary, then*

$$\sup_{\Phi\in\mathcal{F}} \eta_\Phi(X,Y) = s^*(X,Y).$$

Combining this result with Beigi and Gohari's result [2] yields that for binary X and any $\Phi \in \mathcal{F}$,

$$\rho_{\mathrm{m}}^2(X,Y) \le \eta_\Phi(X,Y) \le s^*(X,Y).$$

Both the lower and upper bounds are sharp. The lower bound is attained by $\Phi(t) = t^2$; the upper bound is attained by $\Phi(t) = t\log t$. Furthermore, if we relax Φ to be twice differentiable and convex such that $\Phi''(1) > 0$ (but $\frac{1}{\Phi''}$ is not necessarily to be concave), then the sharp lower bound on $\eta_\Phi(X,Y)$ is still $\rho_{\mathrm{m}}^2(X,Y)$. However, if we additionally require Φ to be operationally convex, the sharp upper bound is $\max\left\{\rho_{\mathrm{m}}^2(X,Y), \sup_{0<\lambda<1}\eta_{\mathrm{LC}_\lambda}(X,Y)\right\}$ where the LC_λ-divergence with $\mathrm{LC}_\lambda(t) := 1 - \frac{t}{\lambda t+1-\lambda}$ corresponds to the Le Cam divergence with parameter λ [1, Theorem III.6].

If $(X,Y) \sim \mathrm{DSBS}(\rho)$, then (1) turns into

$$\eta_\Phi(U,V) \le \rho^2 \tag{5}$$

for all $\Phi \in \mathcal{F}$. Since $\sup_{\Phi\in\mathcal{F}} \eta_\Phi(U,V) \le s^*(U,V)$, the strongest form of (5) over all choices $\Phi \in \mathcal{F}$ is $s^*(U,V) \le \rho^2$. Hence hypercontractivity inequalities are the best strong data processing inequalities for the DSBS in the sense that any other strong data processing inequalities are implied by hypercontractivity inequalities.

It is natural to extend Theorem 1 to cases where X is defined on an arbitrary finite alphabet. We pose the following question.

Problem 1. Is it true that $\sup_{\Phi\in\mathcal{F}} \eta_\Phi(X,Y) = s^*(X,Y)$ for any X, Y defined on finite alphabets?

The answer is yes, if one can show that for any X, Y defined on finite alphabets, given $f \in \widetilde{\mathcal{F}}$,

$$\sup_{g\in\mathcal{G}} \frac{H_g(\mathbb{E}[f|Y])}{H_g(f)}$$

is attained by $g(t) = t$ or $g(t) = 1$.

II. Proof of Theorem 1

For brevity, given f, we denote

$$\mathbb{H}_f : g \mapsto \frac{H_g(\mathbb{E}[f|Y])}{H_g(f)}.$$

We only need to prove that

$$\sup_{f\in\widetilde{\mathcal{F}}} \sup_{g\in\mathcal{G}} \mathbb{H}_f(g) \le s^*(X,Y).$$

To prove this, it suffices to prove the following theorem.

Theorem 2. *For a pair of binary r.v.'s (X,Y) and a given $f \in \widetilde{\mathcal{F}}$, $\sup_{g\in\mathcal{G}} \mathbb{H}_f(g)$ is attained by $g(t) = t$ or $g(t) = 1$.*

Assuming that this theorem holds, we have

$$\begin{aligned}\sup_{f\in\widetilde{\mathcal{F}}} \sup_{g\in\mathcal{G}} \mathbb{H}_f(g) &\le \sup_{f\in\widetilde{\mathcal{F}}} \max\left\{\mathbb{H}_f(t\mapsto t), \mathbb{H}_f(t\mapsto 1)\right\}\\ &= \max\left\{s^*(X,Y), \rho_{\mathrm{m}}^2(X,Y)\right\}\\ &= s^*(X,Y).\end{aligned}$$

That is, Theorem 1 follows. We next prove Theorem 2.

For a pair of binary r.v.'s (X,Y), denote $u_i := f(x_i), p_i := P_X(x_i), i = 0,1$ and $v_j := \mathbb{E}[f(X)|Y = y_j], q_i := P_Y(y_j), 0 \le j \le |\mathcal{Y}|-1$. WLOG, we assume $u_0 \le v_0 \le \ldots \le v_{|\mathcal{Y}|-1} \le u_1$. Then we can write $\mathbb{H}_f(g)$ as

$$\mathbb{H}_f(g) = \frac{\sum_{j=0}^{|\mathcal{Y}|-1} q_j \Phi(v_j)}{p_0\Phi(u_0) + p_1\Phi(u_1)}. \tag{6}$$

We first make the following claim.

Claim 1. For $g \in \mathcal{G}$, define a function $\overline{g}(t) = k(t-1)+1$ for $t \ge 1$ with k satisfying $\overline{\Phi}(u_1) = \Phi(u_1)$. Then we have $\Phi(v) \le \overline{\Phi}(v)$ for any $v \in [1,u_1]$, where $\Phi(x) = \int_1^x \int_1^s \frac{1}{g(t)} dt ds, \overline{\Phi}(x) = \int_1^x \int_1^s \frac{1}{\overline{g}(t)} dt ds$.

Now we prove this claim. We assume that g is not linear, otherwise, this claim follows trivially. By the concavity of g, we know that one of the following cases holds: 1) $g \ge \overline{g}$ on the whole interval $[1,u_1]$; 2) $g \le \overline{g}$ on the whole interval $[1,u_1]$; 3) there is a value $b \in (1,u_1)$ such that $g \ge \overline{g}$ on the interval $[1,b]$ and $g \le \overline{g}$ on the interval $[b,u_1]$. By the assumption $\overline{\Phi}(u_1) = \Phi(u_1)$, we have that only the case 3) could happen. This implies that $h := \frac{1}{\overline{g}} - \frac{1}{g}$ is positive (or zero) on the interval $[1,b]$ and negative (or zero) on the interval $[b,u_1]$. By this, we further have that $s \mapsto \int_1^s h(t)\,dt$ is increasing on the interval

$[1,b]$ and decreasing on the interval $[b,u_1]$. Hence one of the following cases holds: 1) $s \mapsto \int_1^s h(t)\,dt$ is positive on the whole interval $[1,u_1]$; 2) there is a value $b' \in (1,u_1)$ such that $s \mapsto \int_1^s h(t)\,dt$ is positive on the interval $[1,b']$ and negative on the interval $[b',u_1]$. By the assumption $\overline{\Phi}(u_1) = \Phi(u_1)$, i.e., $\int_1^{u_1}\int_1^s h(t)\,dtds = 0$, we have that only the case 2) could happen. Therefore, $x \mapsto \int_1^x \int_1^s h(t)\,dt$ is increasing on the interval $[1,b']$ and decreasing on the interval $[b',u_1]$, and moreover, it is zero at $x = u_1$. It means that $x \mapsto \int_1^x \int_1^s h(t)\,dt$ is positive (or zero) on the whole interval $[1,u_1]$. Hence $\Phi(v) \le \overline{\Phi}(v)$ for any $v \in [1,u_1]$.

By Claim 1, all $\Phi(v_j)$ with $v_j \ge 1, 0 \le j \le |\mathcal{Y}|-1$ are maximized simultaneously by $\overline{g}$ over all $g \in \mathcal{G}$ under the condition that $\Phi(u_1)$ is fixed.

Similarly, one can show that all $\Phi(v_j)$ with $v_j \le 1, 0 \le j \le |\mathcal{Y}|-1$ are maximized simultaneously by $\widehat{g}$ over all $g \in \mathcal{G}$ under the condition that $\Phi(u_0)$ is fixed, where $\widehat{g}(t) := k'(t-1)+1$ for $t \le 1$ with k' satisfying $\widehat{\Phi}(u_0) = \Phi(u_0)$.

We choose $\ddot{g}(t) = \begin{cases} \overline{g}(t) & t \ge 1 \\ \widehat{g}(t) & t \le 1 \end{cases}$. Then by (6), $\mathbb{H}_f(\ddot{g}) \ge \mathbb{H}_f(g)$, which implies that the optimal g^* attaining $\sup_{g\in\mathcal{G}} \mathbb{H}_f(g)$ is either linear or two-pieces linear with turning point at $t=1$. Obviously, $\sup_{f\in\widetilde{\mathcal{F}}} \mathbb{H}_f(g)$ induced by linear g is exactly $s^*(X,Y)$. Hence it suffices to only consider the two-pieces linear case. We next prove that for a two-pieces linear function g with turning point at $t=1$, the induced $\mathbb{H}(g)$ is no larger than the $\mathbb{H}(\check{g})$ induced by a linear function $\check{g}$. To this end, we need the following lemmas.

For a function $g \in \mathcal{G}$, denote $\overline{g} \in \mathcal{G}$ as a linear function $t \mapsto k(t-1)+1$ with $0 \le k \le 1$ such that $\overline{g} \ge g$. Let $\Phi, \overline{\Phi}$ be respectively the twice-integral of $1/g, 1/\overline{g}$. By definition, any $g \in \mathcal{G}$ is nonnegative and nondecreasing, and g' is nonincreasing. Hence $\Phi''(t) \ge 0$ and $\Phi'''(t) = (1/g)' = -g'/g^2 \le 0$. Moreover, $g'(t) \ge k$ for $0 \le t \le 1$ and $g'(t) \le k$ for $t \ge 1$. Obviously, it also holds that $\overline{\Phi}''(t) \ge 0$ and $\overline{\Phi}'''(t) \le 0$.

Lemma 4. *$\frac{\Phi''(t)}{\overline{\Phi}''(t)} = \frac{\overline{g}(t)}{g(t)}$ is decreasing for $t \le 1$ and increasing for $t \ge 1$ and $\frac{\overline{g}(1)}{g(1)} = 1$.*

Proof: Observe that

$$\begin{aligned}\left[\frac{\overline{g}(t)}{g(t)}\right]' &= \frac{\overline{g}'(t)\,g(t) - g'(t)\,\overline{g}(t)}{[g(t)]^2} \\ &= \frac{\overline{g}(t)}{g(t)}\left[\frac{\overline{g}'(t)}{\overline{g}(t)} - \frac{g'(t)}{g(t)}\right]\end{aligned}$$

For $t > 1$,

$$\frac{\overline{g}(t)}{g(t)} = \frac{1+\int_1^t \overline{g}'(t)\,ds}{1+\int_1^t g'(t)\,ds} \le \max\left\{1,\ \sup_{1<s<t}\frac{\overline{g}'(s)}{g'(s)}\right\}$$

Since $\overline{g}' = k$ is constant, and g' is nonincreasing, we have that $\frac{\overline{g}'(s)}{g'(s)}$ is nondecreasing. Hence

$$\sup_{1<s<t}\frac{\overline{g}'(s)}{g'(s)} = \frac{\overline{g}'(t)}{g'(t)} \ge 1,$$

which further implies that for $t > 1$,

$$\frac{\overline{g}(t)}{g(t)} \le \frac{\overline{g}'(t)}{g'(t)}.$$

Hence $\left[\frac{\overline{g}(t)}{g(t)}\right]' \ge 0$ for $t > 1$.

Similarly, one can prove that $\left[\frac{\overline{g}(t)}{g(t)}\right]' \le 0$ for $t < 1$. ■

Lemma 5. *$\frac{\Phi'(t)}{\overline{\Phi}'(t)}$ is decreasing for $t \le 1$ and increasing for $t \ge 1$ and $\frac{\Phi'(1)}{\overline{\Phi}'(1)} = 1$.*

Proof: Observe that

$$\begin{aligned}\left[\frac{\Phi'(t)}{\overline{\Phi}'(t)}\right]' &= \frac{\Phi''(t)\,\overline{\Phi}'(t) - \overline{\Phi}''(t)\,\Phi'(t)}{\left[\overline{\Phi}'(t)\right]^2} \\ &= \frac{\overline{\Phi}''(t)}{\overline{\Phi}'(t)}\left[\frac{\Phi''(t)}{\overline{\Phi}''(t)} - \frac{\Phi'(t)}{\overline{\Phi}'(t)}\right]\end{aligned}$$

and $\frac{\Phi''(t)}{\overline{\Phi}''(t)} = \frac{\overline{g}(t)}{g(t)}$.

For $t > 1$,

$$\frac{\Phi'(t)}{\overline{\Phi}'(t)} = \frac{\int_1^t \frac{1}{g(s)} ds}{\int_1^t \frac{1}{\overline{g}(s)} ds} \leq \sup_{1<s<t} \frac{\frac{1}{g(s)}}{\frac{1}{\overline{g}(s)}} = \frac{\overline{g}(t)}{g(t)} = \frac{\Phi''(t)}{\overline{\Phi}''(t)}$$

and $\overline{\Phi}'(t), \overline{\Phi}''(t) > 0$. Hence $\left[\frac{\Phi'(t)}{\overline{\Phi}'(t)}\right]' \geq 0$.

For $t < 1$,

$$\frac{\Phi'(t)}{\overline{\Phi}'(t)} = \frac{\int_t^1 \frac{1}{g(s)} ds}{\int_t^1 \frac{1}{\overline{g}(s)} ds} \leq \sup_{t<s<1} \frac{\frac{1}{g(s)}}{\frac{1}{\overline{g}(s)}} = \frac{\overline{g}(t)}{g(t)} = \frac{\Phi''(t)}{\overline{\Phi}''(t)}$$

and $\overline{\Phi}'(t) < 0, \overline{\Phi}''(t) > 0$. Hence $\left[\frac{\Phi'(t)}{\overline{\Phi}'(t)}\right]' \leq 0$. ■

Lemma 6. *$\frac{\Phi(t)}{\overline{\Phi}(t)}$ is decreasing for $t \leq 1$ and increasing for $t \geq 1$ and $\frac{\Phi(1)}{\overline{\Phi}(1)} = 1$.*

Proof: Observe that

$$\left[\frac{\Phi(t)}{\overline{\Phi}(t)}\right]' = \frac{\Phi'(t)\overline{\Phi}(t) - \overline{\Phi}'(t)\Phi(t)}{\overline{\Phi}^2(t)}$$

and

$$\begin{aligned}\left[\Phi'(t)\overline{\Phi}(t) - \overline{\Phi}'(t)\Phi(t)\right]' &= \Phi''(t)\overline{\Phi}(t) - \overline{\Phi}''(t)\Phi(t) \\ &= \overline{\Phi}(t)\overline{\Phi}''(t)\left[\frac{\Phi''(t)}{\overline{\Phi}''(t)} - \frac{\Phi(t)}{\overline{\Phi}(t)}\right]\end{aligned}$$

For $t > 1$,

$$\begin{aligned}\frac{\Phi''(t)}{\overline{\Phi}''(t)} &= \frac{\overline{g}(t)}{g(t)}, \\ \frac{\Phi(t)}{\overline{\Phi}(t)} &= \frac{\int_1^r \int_1^s \frac{1}{g(t)} dt ds}{\int_1^r \int_1^s \frac{1}{\overline{g}(t)} dt ds} \leq \sup_{1<s<t} \frac{\frac{1}{g(s)}}{\frac{1}{\overline{g}(s)}} = \frac{\overline{g}(t)}{g(t)},\end{aligned}$$

which implies $\left[\Phi'(t)\overline{\Phi}(t) - \overline{\Phi}'(t)\Phi(t)\right]' \geq 0$. Hence we have

$$\Phi'(t)\overline{\Phi}(t) - \overline{\Phi}'(t)\Phi(t) \geq \Phi'(1)\overline{\Phi}(1) - \overline{\Phi}'(1)\Phi(1) = 0$$

That is, $\left[\frac{\Phi(t)}{\overline{\Phi}(t)}\right]' \geq 0$.

For $t < 1$, similarly, we have

$$\left[\Phi'(t)\overline{\Phi}(t) - \overline{\Phi}'(t)\Phi(t)\right]' \geq 0$$

Hence

$$\Phi'(t)\overline{\Phi}(t) - \overline{\Phi}'(t)\Phi(t) \leq \Phi'(1)\overline{\Phi}(1) - \overline{\Phi}'(1)\Phi(1) = 0$$

That is, $\left[\frac{\Phi(t)}{\overline{\Phi}(t)}\right]' \leq 0$. ■

Lemma 7. *For $a_i, b_i, c_i, d_i, i = 0, 1$ such that $b_i > d_i > 0$ and*

$$\frac{a_i}{b_i} \leq \frac{c_i}{d_i}, \tag{7}$$

we have

$$\frac{a_0 + a_1}{b_0 + b_1} \leq \max\left\{\frac{a_0 + c_1}{b_0 + d_1}, \frac{c_0 + a_1}{d_0 + b_1}\right\}.$$

Proof: Case of $\frac{a_0}{b_0} \geq \frac{a_1}{b_1}$: By (7),

$$\frac{a_1 - c_1}{b_1 - d_1} \leq \frac{a_1}{b_1}$$

If $\frac{a_0}{b_0} \geq \frac{a_1}{b_1}$, then

$$\frac{a_1 - c_1}{b_1 - d_1} \leq \frac{a_0}{b_0}$$

That is,

$$b_0\left(a_1-c_1\right)\leq a_0\left(b_1-d_1\right) \tag{8}$$

By (7) and (8),

$$a_0b_0+b_0\left(a_1-c_1\right)+a_1d_1\leq a_0b_0+a_0\left(b_1-d_1\right)+c_1b_1$$

That is,

$$\frac{a_0+a_1}{b_0+b_1}\leq\frac{a_0+c_1}{b_0+d_1}$$

Case of $\frac{a_0}{b_0}<\frac{a_1}{b_1}$: Similarly, one can show that if $\frac{a_0}{b_0}<\frac{a_1}{b_1}$, then

$$\frac{a_0+a_1}{b_0+b_1}\leq\frac{c_0+a_1}{d_0+b_1}$$

■

Let $g_1\left(t\right):=k_1\left(t-1\right)+1$, $g_2\left(t\right):=k_2\left(t-1\right)+1$, and $g\left(t\right):=\begin{cases}g_1\left(t\right) & t\leq 1\\ g_2\left(t\right) & t>1\end{cases}$, where $k_1\geq k_2$. We next show the following lemma.

Lemma 8. *Let* $U:=f\left(X\right),V:=\mathbb{E}[f\left(X\right)|Y]$. *Then*

$$\frac{H_\Phi(V)}{H_\Phi(U)}\leq\max\left\{\frac{H_{\Phi_1}(V)}{H_{\Phi_1}(U)},\frac{H_{\Phi_2}(V)}{H_{\Phi_2}(U)}\right\}$$

where Φ,Φ_1,Φ_2 *are respectively the twice-integrals of* $1/g,1/g_1,1/g_2$.

Proof: For binary sources, denote $u_i:=f\left(x_i\right),i=0,1$ and $v_j:=\mathbb{E}[f\left(X\right)|Y=y_j],j=0,1,...,N-1$. WLOG, we assume $u_0\leq v_0\leq...\leq v_{N_0-1}\leq 1\leq v_{N_0}\leq...\leq v_{N-1}\leq u_1$ for some $1\leq N_0\leq N$. By Lemma 6,

$$\begin{aligned}1\leq\frac{\Phi\left(v_j\right)}{\Phi_2\left(v_j\right)}\leq\frac{\Phi\left(u_0\right)}{\Phi_2\left(u_0\right)}\text{ for }j\leq N_0-1,\\ 1\leq\frac{\Phi\left(v_j\right)}{\Phi_1\left(v_j\right)}\leq\frac{\Phi\left(u_1\right)}{\Phi_1\left(u_1\right)}\text{ for }j\geq N_0.\end{aligned}$$

That is,

$$\begin{aligned}\frac{\Phi\left(v_j\right)}{\Phi\left(u_0\right)}\leq\frac{\Phi_2\left(v_j\right)}{\Phi_2\left(u_0\right)}\text{ for }j\leq N_0-1,\\ \frac{\Phi\left(v_j\right)}{\Phi\left(u_1\right)}\leq\frac{\Phi_1\left(v_j\right)}{\Phi_1\left(u_1\right)}\text{ for }j\geq N_0,\end{aligned}$$

which further imply

$$\begin{aligned}\frac{\sum_{j=0}^{N_0-1}P_Y\left(y_j\right)\Phi\left(v_j\right)}{P_X\left(x_0\right)\Phi\left(u_0\right)}\leq\frac{\sum_{j=0}^{N_0-1}P_Y\left(y_j\right)\Phi_2\left(v_j\right)}{P_X\left(x_0\right)\Phi_2\left(u_0\right)},\\ \frac{\sum_{j=N_0}^{N-1}P_Y\left(y_j\right)\Phi\left(v_j\right)}{P_X\left(x_1\right)\Phi\left(u_1\right)}\leq\frac{\sum_{j=N_0}^{N-1}P_Y\left(y_j\right)\Phi_1\left(v_j\right)}{P_X\left(x_1\right)\Phi_1\left(u_1\right)}.\end{aligned}$$

By Lemma 7,

$$\frac{H_\Phi(V)}{H_\Phi(U)}\leq\max\left\{\frac{H_{\Phi_1}(V)}{H_{\Phi_1}(U)},\frac{H_{\Phi_2}(V)}{H_{\Phi_2}(U)}\right\}.$$

■

Lemma 8 implies that for a given f and a two-pieces linear function g, the induced $\mathbb{H}_f\left(g\right)$ is no larger than the one induced by a linear function g. This implies Theorem 1, since the supremum of $\mathbb{H}_f\left(g\right)$ over all $f\in\widetilde{\mathcal{F}}$ for a linear function g, is equal to the supremum of $\mathbb{H}_f\left(g\right)$ over all $f\in\widetilde{\mathcal{F}}$ for function $g\left(t\right)=t$. The latter supremum is exactly $s^*(X,Y)$.

Appendix A
Proof of Lemma 3

WLOG, one can restrict Φ to be a function such that

$$\Phi(1) = 0 \tag{9}$$
$$\Phi'(1) = 0 \tag{10}$$
$$\Phi''(1) = 1, \tag{11}$$

since for fixed f, $\frac{H_\Phi(\mathbb{E}[f|Y])}{H_\Phi(f)}$ does not change if we replace $\Phi(t)$ with $\alpha\Phi(t) + \beta t + \gamma$ for $\alpha > 0, \beta, \gamma \in \mathbb{R}$. Obviously, in this setting, Φ is nonnegative, and uniquely determined by Φ''. Since 1) Φ'' is positive on $\mathbb{R}_{>0}$, 2) $\frac{1}{\Phi''(1)} = 1$, and 3) $\frac{1}{\Phi''}$ is concave on $\mathbb{R}_{>0}$, we have

$$1 \le \frac{1}{\Phi''(t)} \le t, \forall t \in [1, \infty)$$
$$t \le \frac{1}{\Phi''(t)} \le 1, \forall t \in (0, 1)$$

i.e.,

$$\frac{1}{t} \le \Phi''(t) \le 1, \forall t \in [1, \infty)$$
$$1 \le \Phi''(t) \le \frac{1}{t}, \forall t \in (0, 1)$$

Denote $A = \{f(x_1), ..., f(x_M), \mathbb{E}[f|y_1], .., \mathbb{E}[f|y_N]\}$. By definition, $0 < a_{\min} := \min A < a_{\max} := \max A < +\infty$. Observe that $\frac{1}{a_{\max}} \le \Phi''(t) \le \frac{1}{a_{\min}}$ for all $t \in [a_{\min}, a_{\max}]$.

Obviously, $g := \frac{1}{\Phi''}$ is positive, concave, continuous, increasing on $\mathbb{R}_{>0}$. Hence

$$R_g(x_1, x_2) := \frac{g(x_1) - g(x_2)}{x_1 - x_2}$$

is nonnegative, and monotonically non-increasing in x_1 for every fixed x_2, (or vice versa). Hence for $x_1, x_2 \in [a_{\min}, a_{\max}]$,

$$R_g(x_1, x_2) \le \frac{g(a_{\min}) - g(x_2)}{a_{\min} - x_2} \le \frac{g(a_{\min}) - g(0)}{a_{\min} - 0} \le \frac{g(a_{\min})}{a_{\min}} \le \frac{1}{a_{\min}}.$$

Hence $R_g(x_1, x_2)$ is uniformly bounded on $[a_{\min}, a_{\max}]$ for all functions $g = \frac{1}{\Phi''}$ with $\Phi \in \mathcal{F}$. Observe that

$$R_{\Phi''}(x_1, x_2) = \frac{\Phi''(x_1) - \Phi''(x_2)}{x_1 - x_2} = \frac{\frac{1}{g(x_1)} - \frac{1}{g(x_2)}}{x_1 - x_2} = -\frac{R_g(x_1, x_2)}{g(x_1)g(x_2)}.$$

Hence

$$-\frac{1}{a_{\min}^3} \le R_{\Phi''}(x_1, x_2) \le 0.$$

which implies that $R_{\Phi''}(x_1, x_2)$ is also uniformly bounded on $[a_{\min}, a_{\max}]$ for all functions $\Phi \in \mathcal{F}$.

Given f, let $\{\Phi_n\}$ be a sequence of functions that attains $\sup_{\Phi \in \mathcal{F}} \frac{H_\Phi(\mathbb{E}[f|Y])}{H_\Phi(f)}$. Since $\Phi_n \in \mathcal{F}$, $R_{g_n}(x_1, x_2)$ with $g_n := \frac{1}{\Phi_n''}$ is uniformly bounded on $[a_{\min}, a_{\max}]$. This means that the sequence $\{g_n\}$ is uniformly equicontinuous. By the ArzelÃ –Ascoli theorem, there exists a subsequence $\{g_{n_k}\}_{k \in \mathbb{N}}$ that converges uniformly on $[a_{\min}, a_{\max}]$. Let $\widehat{g}$ be the limit of this subsequence. Then $\frac{1}{g_{n_k}} \to \frac{1}{\widehat{g}}$ and $\Phi_{n_k} \to \widehat{\Phi}$ uniformly on $[a_{\min}, a_{\max}]$, where $\widehat{\Phi}$ is the twice-integral of $\frac{1}{\widehat{g}}$ satisfying (9)-(11). This means

$$\sup_{\Phi \in \mathcal{F}} \frac{H_\Phi(\mathbb{E}[f|Y])}{H_\Phi(f)} = \lim_{k \to \infty} \frac{H_{\Phi_{n_k}}(\mathbb{E}[f|Y])}{H_{\Phi_{n_k}}(f)} = \frac{H_{\widehat{\Phi}}(\mathbb{E}[f|Y])}{H_{\widehat{\Phi}}(f)}.$$

We next construct an extension $\widetilde{\Phi}$ of $\widehat{\Phi}$ to the domain $\mathbb{R}_{>0}$ such that $\widetilde{\Phi} \in \mathcal{F}$. Obviously, $\widehat{g}$ is concave, and it satisfies $a_{\min} \le \widehat{g}(t) \le a_{\max}$ for all $t \in [a_{\min}, a_{\max}]$. Furthermore, for all $x_1, x_2 \in [a_{\min}, a_{\max}]$,

$$0 \le R_{\widehat{g}}(x_1, x_2) \le \frac{\widehat{g}(a_{\min})}{a_{\min}}. \tag{12}$$

We now construct an extension of $\widehat{g}$ as

$$\widetilde{g}(t) = \begin{cases} \widehat{g}(t) & t \in [a_{\min}, a_{\max}] \\ \frac{\widehat{g}(a_{\min})}{a_{\min}} t & t \in (0, a_{\min}) \\ \widehat{g}(a_{\max}) & t \in (a_{\max}, +\infty) \end{cases}$$

Obviously, $\widetilde{g} > 0$. Moreover, by the definition of $\widetilde{g}$, inequalities in (12), and the fact that $\widehat{g}$ is concave, we have that $\widetilde{g}$ is concave. Define $\widetilde{\Phi}$ as the twice-integral of $\frac{1}{\widetilde{g}}$ satisfying (9)-(11). Then obviously, $\widetilde{\Phi} \in \mathcal{F}$ and

$$\frac{H_{\widetilde{\Phi}}(\mathbb{E}[f|Y])}{H_{\widetilde{\Phi}}(f)} = \frac{H_{\widehat{\Phi}}(\mathbb{E}[f|Y])}{H_{\widehat{\Phi}}(f)} = \sup_{\Phi \in \mathcal{F}} \frac{H_{\Phi}(\mathbb{E}[f|Y])}{H_{\Phi}(f)}.$$

This implies that given f, $\sup_{\Phi \in \mathcal{F}} \frac{H_{\Phi}(\mathbb{E}[f|Y])}{H_{\Phi}(f)}$ is attained by $\widetilde{\Phi}$.

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